

RUTURAJ BHASKAR NAWALE

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PROFILE SUMMARY

AI Engineer with hands-on experience in designing and developing AI-powered enterprise applications across Computer Vision, OCR, Deep Learning, LLMs, Fraud Detection, Deepfake Detection, Merchant Intelligence, and Banking Analytics. Proficient in Python, FastAPI, PyTorch, OpenCV, and REST APIs, with expertise in building end-to-end AI pipelines, scalable backend services, risk scoring systems, and explainable AI solutions. Experienced in integrating machine learning models, LLMs, and business intelligence workflows to deliver production-ready AI systems for real-world enterprise applications.

SKILLS

- **AI & Machine Learning:** Computer Vision, Deep Learning, OCR, NLP, LLMs, Generative AI, Prompt Engineering, Fraud Detection, Deepfake Detection, Predictive Analytics, Risk Scoring
- **Frameworks & Libraries:** PyTorch, OpenCV, PaddleOCR, TrOCR, LayoutLMv3, InsightFace, ArcFace, EfficientNet, Vision Transformer (ViT), XGBoost, Prophet, Hugging Face
- **Backend & Data:** Python, FastAPI, REST APIs, JSON, BeautifulSoup, Selenium, Playwright, Pandas, Polars
- **Databases & Tools:** MSSQL Server, PostgreSQL, SQLite

EXPERIENCE

AI Engineer:- Jode Technologies Pvt. Ltd. | May 2026-Present

- Developed production-grade AI platforms for merchant KYC verification, deepfake detection, merchant profiling, and banking reconciliation using Python, FastAPI, PyTorch, OpenCV, OCR, LLMs, and Machine Learning
- Designed scalable end-to-end AI pipelines integrating computer vision, OCR, document understanding, fraud detection, predictive analytics, risk scoring, REST APIs, and explainable AI for intelligent automation

Web Developer (Intern):- ReaEspresso PropTech Pvt. Ltd | April 2025-June 2025

- Contributed to the development and optimisation of production web pages in a collaborative, agile environment
- Implemented responsive UI components aligned with business and performance requirements
- Worked closely with designers and developers to deliver client-facing features within timelines

PROJECTS

Merchant Profiling & Business Intelligence Platform

Python, FastAPI, BeautifulSoup, Selenium, Playwright, Pandas, JSON, REST APIs, LLMs

- Developed an AI-powered merchant profiling platform that builds a 360° business intelligence profile by accepting a domain, GSTIN, CIN, or company name, performing multi-source data collection, website crawling, GST/CIN validation, domain intelligence, technology fingerprinting, social media discovery, security analysis, and automated risk scoring.
- Built a scalable data extraction and enrichment pipeline using BeautifulSoup, Selenium, Playwright, and REST APIs to extract company information, contact details, GST/CIN records, WHOIS/DNS/SSL intelligence, technology stack, and public business data, consolidating all information into standardized JSON.

AI 5V Reconciliation & Banking Intelligence Platform

Python, FastAPI, SQL Server, Polars, XGBoost, Prophet, ARIMA, Isolation Forest, REST APIs, LLMs

- Developed an enterprise AI-powered reconciliation platform for banking and payment systems (UPI, NPCI, CBS) by designing end-to-end pipelines for data ingestion, transaction reconciliation, anomaly detection, forecasting, and AI-driven operational intelligence.
- Designed five interactive intelligence dashboards—Global, Transaction, Account (360° Profile), Predictive, and Resolution Intelligence—to deliver executive KPIs, transaction lifecycle tracking, reconciliation exception management, AI-powered forecasting, root cause analysis, and operational decision support.

Merchant Verification & Fraud Detection System

Python, FastAPI, PaddleOCR, TrOCR, LayoutLMv3, OpenCV, InsightFace, ArcFace, XGBoost, PyTorch, Transformers, JSON

- Developed an AI-powered merchant KYC verification platform that automates digital onboarding through an end-to-end pipeline including document upload, image quality assessment, document classification, OCR (PaddleOCR + TrOCR), structured JSON extraction, face verification, fraud detection, risk scoring, and automated approval/review/rejection.
- Built intelligent document and identity verification modules using LayoutLMv3, InsightFace, ArcFace, OpenCV, and XGBoost for document understanding, key-value extraction, biometric face matching, image forensics, metadata validation, and detection of tampered documents, fake identities, and OCR inconsistencies.

Deepfake Detection & Media Authenticity Verification System

Python, FastAPI, PyTorch, OpenCV, EfficientNet, Vision Transformer (ViT), XceptionNet, RetinaFace, InsightFace, LLMs

- Developed an end-to-end AI-powered deepfake detection platform for image, video, and live-stream verification by designing a pipeline comprising media validation, frame extraction, face detection and alignment, feature extraction, deep learning inference, artifact analysis, metadata validation, temporal consistency checks, risk scoring, and automated genuine/suspicious/deepfake classification.
- Built a multi-model deep learning detection engine using EfficientNet, Vision Transformer (ViT), XceptionNet, RetinaFace, InsightFace, OpenCV, and ensemble learning to identify AI-generated media through facial artifacts, compression traces, and temporal anomalies.

REASEARCH & PUBLICATIONS

Consumer Sentiment Insights on Smart Ring Using Supervised Classifiers

- Conducted sentiment analysis using multiple ML models and comparative evaluation
- Presented findings at National Conference on Robotics & AI (March 2025)

EDUCATION

Changu Kana Thakur College of arts, commerce and science Bachelor's in Computer Science (CS); SGPA: 9.40	Navi-Mumbai, Maharashtra, India <i>June 2023- July 2026</i>
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Navi-Mumbai, Maharashtra, India
June 2023- July 2026